

ASA Overdose



Section I: Scenario Demographics

Scenario Title:	ASA Overdose		
Date of Development:	30/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To allow learners to practice management of a rare presentation with critical steps in its management.
CRM Objectives:	Clearly communicate to team the importance of findings as well as management steps.
Medical Objectives:	1) Recognize the importance and severity of a salicylate overdose. 2) Demonstrate appropriate initial investigations and management of a salicylate overdose. 3) Recognize the significance of a change in mental status in the context of a salicylate overdose.

Case Summary: Brief Summary of Case Progression and Major Events

22 year-old female presents saying she just ingested 60 tablets of ASA because she wants to die. Her mom found her while she was finishing the bottle of 325mg tabs approximately 60 minutes ago and called EMS. The patient is complaining of nausea and tinnitus and is tachypneic. The team should consider activated charcoal and alkalinize the urine. If they do not initiate treatments, they will receive a critical VBG showing a mixed respiratory alkalosis and metabolic acidosis. The patient will then become somnolent. The team will be expected to check her blood sugar and call for dialysis. They will also need to intubate and recognize the need to hyperventilate and dialyze.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



ASA Overdose



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Select most important dimension(s)			
Confederates			
Brief Description of Role			
None.			
**Could add a difficult mother for seniors			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required. Could have empty ASA bottle that family members brought in.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	20 min

ASA Overdose



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Cassandra Acton is a 22 year-old female presents saying she just ingested 60 tablets of ASA because she wants to die. Her mom found her while she was finishing the bottle of 325mg tabs approximately 60 minutes ago and called EMS. The patient is complaining of nausea and tinnitus.

B. Patient Profile and History

Patient Name: Cassandra Acton	Age: 22	Weight: 70kg
Gender: ☐ M ☒ F	Code Status: Full	
Chief Complaint: Overdose		
History of Presenting Illness: Ingested 60 tablets of ASA approximately 90 minutes ago. Found by her mom while doing so. Intent was to die.		
Past Medical History:	None	Medications: None

Allergies: None

Social History: Lives with mom and stepdad as well as stepsister. **(Recent marriage of mom & stepdad.)**

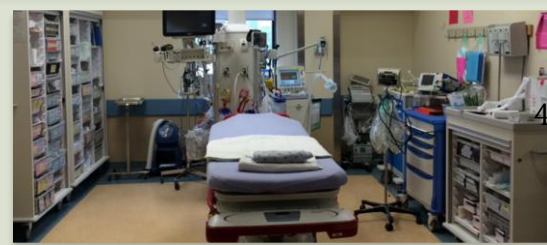
Review of Systems:	CNS:	Ongoing suicidal ideation.		
	HEENT:	Tinnitus.		
	CVS:	No CP, no palps.		
	RESP:	Mildly SOB.		
	GI:	Nausea ongoing.		
	GU:	Nil.		
	MSK:	Nil.	INT:	Nil.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 115/min	BP: 100/60	RR: 30/min
O ₂ SAT: 97%		
Rhythm: Sinus tach	T: 37.6°C	Glucose: 5.3 mmol/L
GCS: 15 (E4 V5 M6)		
General Status: Very tachypneic but otherwise well-seeming.		
CNS:	PERLA. Normal EOM.	
HEENT:	No signs HI.	
CVS:	No murmur.	
RESP:	Tachypneic, but no increased WOB. GAEB with no adventitious sounds.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	No signs trauma.	SKIN: No signs cutting.



ASA Overdose



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 115/min BP: 100/60 RR: 30/min O ₂ SAT: 97% T: 37.6°C	Patient awake, answering questions. Complaining of nausea and tinnitus.	<u>Learner Actions</u> - ☐ IV, monitors - ☐ Tox bloodwork including VBG, osmols, ASA, APAP, EtOH - ☐ ECG - ☐ Cap sugar: 5.3 - ☐ Call poison control - ☐ Activated charcoal - ☐ IV NS 1L bolus	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 4 min → 2. VBG back
2. VBG Back Vitals unchanged	**State starts with giving team critical VBG results**	<u>Learner Actions</u> - ☐ Alkalinize urine - ☐ Call poison control if haven't - ☐ Insert foley - ☐ Bolus IV fluid - ☐ Correct K depletion	<u>Modifiers</u> <u>Triggers</u> - 7 min → 3. Somnolent - Urine alkalinization started → 3. Somnolent
3. Somnolent RR → 26 O ₂ SAT → 93%	Patient now unresponsive. GCS 8 (E2 V2 M4)	<u>Learner Actions</u> - ☐ Cap sugar: 3.8 - ☐ Give 1 amp D50 iv - ☐ Prepare for intubation - ☐ Elevate HOB - ☐ Call for dialysis	<u>Modifiers</u> - No cap sugar by 2 min into state → patient seizes - O ₂ applied → O ₂ SAT to 95% <u>Triggers</u> - Intubation → 5. Post-Intubation - 12 min → 6. ASA Back
4. Seizure HR → 150 BP → 160/90	Patient actively seizing	<u>Learner Actions</u> - ☐ Check cap sugar: 3.8 - ☐ Give 1 amp D50 iv - ☐ Continue urine alkalinization	<u>Modifiers</u> <u>Triggers</u> - D50 given → 6. ASA Back
5. Post-Intubation HR → 105 BP → 95/55 O ₂ SAT → 100% RR → vent settings indicated by team	**Give team ASA level at onset of this state**	<u>Learner Actions</u> - ☐ Set ventilator with high RR - ☐ Continue urine alkalinization - ☐ Call ICU - ☐ Call Nephrology for dialysis - ☐ Post-intubation CXR - ☐ Administer sedation - ☐ Elevate HOB	END CASE PRN with ICU arriving.
6. ASA Back Same as previous	**Give ASA level at onset of this state**	<u>Learner Actions</u> - ☐ Call Nephrology for dialysis - ☐ Continue urine alkalinization - ☐ Call ICU - ☐ Call poison control if haven't yet	END CASE PRN with ICU arriving.

ASA Overdose

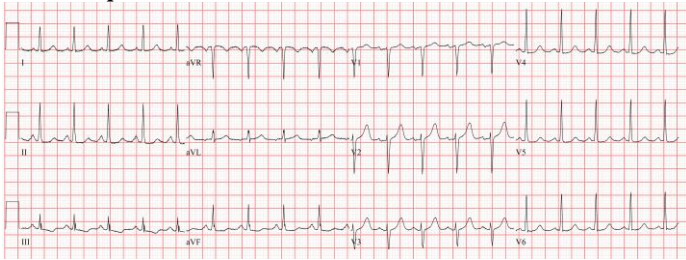


Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 135	K: 3.2	Cl: 100	HCO ₃ : 8	BUN: 7	Cr: 80	Glu: 5.0
VBG		pH: 7.30	PCO ₂ : 18	PO ₂ : 40	HCO ₃ : 12	Lactate: 4
ASA: 7.5mmol/L						

Images (ECGs, CXRs, etc.)

ECG at presentation



Source:

<http://en.ecgpedia.org/wiki/File:Sinustachycardia.jpg>

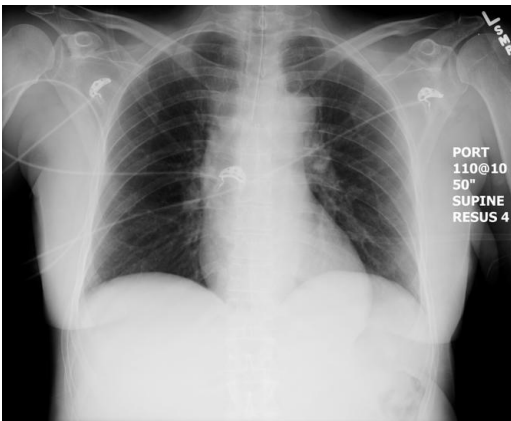
CXR at presentation



Source:

<http://radiologypics.com/2013/01/25/normal-female-chest-radiograph/>

Post-intubation CXR



Source: <https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To allow learners to practice management of a rare presentation with critical steps in its management.
CRM Objectives:	Clearly communicate the importance of findings and management to the team.
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Sample Questions for Debriefing	
1) What are the clinical features of ASA overdose? 2) What are the indications for urine alkalinization? How do you do it? 3) Why do you think the patient became more comatose? What causes of this are important to consider in an ASA overdose? 4) What are the indications for dialysis in an ASA overdose? 5) What acid-base disturbances are seen in an ASA overdose? How does the anion gap change? 6) Do you feel that your team was on the same page about the severity of this overdose? Do you feel you were all aware of what needed to be done? If not, do you think the leader managed to show why? 7) Why is intubation in an ASA overdose worrisome? What can you do pre and post intubation to mitigate these risks?	
Key Moments	
Recognition of metabolic acidosis and initiation of urine alkalinization	
Management of altered LOC	
Post-intubation management (calling consultants urgently)	